Experiment No.:5

Simulation of a Smart Hospital Live Patient Monitoring System Using Python (Google Colab)

Aim

To develop and simulate a Smart Hospital Live Patient Monitoring System using Python in Google Colab that continuously generates and monitors the vital signs of multiple patients, displays the data in real time through live tables and graphs, automatically identifies emergency cases based on predefined medical thresholds, and provides instant emergency alerts.

Objective

The objective of this experiment is to simulate a smart hospital patient monitoring system using Python in Google Colab. The system continuously monitors the vital signs of 25 patients by generating simulated values for body temperature, heart rate, and blood oxygen saturation (SpO_2) at regular intervals of five seconds. The monitored data are displayed in a live dashboard containing continuously updating tables and graphical visualizations for real-time observation. The system analyses each patient's health condition by comparing the generated values with predefined threshold limits and classifies them as either NORMAL or EMERGENCY. Whenever abnormal vital signs are detected, the system immediately displays an emergency alert along with the patient's details. This experiment demonstrates how live patient monitoring, automated health status classification, emergency detection, and real-time visualization can be implemented through simulation using Python in Google Colab.

Software Requirements

Google Collab

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from IPython.display import clear_output, display

import random

```python
import time

# Number of patients
total_patients = 25

patients = []
temperature = []
heart_rate = []
spo2 = []
status = []

for i in range(1, total_patients + 1):

    # Generate random values
    temp = round(random.uniform(34.5, 39.5), 1)
    hr = random.randint(50, 120)
    oxygen = random.randint(90, 100)

    # Check status
    if temp < 35 or temp > 38 or hr < 60 or hr > 100 or oxygen < 95:
        stat = "EMERGENCY"
    else:
        stat = "NORMAL"

    patients.append(f"P{i:02d}")
    temperature.append(temp)
```

```
heart_rate.append(hr)
spo2.append(oxygen)
status.append(stat)

df = pd.DataFrame({
    "Patient": patients,
    "Temperature (°C)": temperature,
    "Heart Rate (BPM)": heart_rate,
    "SpO₂ (%)": spo2,
    "Status": status
})

clear_output(wait=True)

print("SMART HOSPITAL LIVE PATIENT MONITORING SYSTEM")
print("="*50)

display(df)

emergency = df[df["Status"] == "EMERGENCY"]

print("\nEmergency Patients")

if len(emergency) > 0:
    display(emergency)
else:
```

```python
    print("No Emergency Cases")

plt.figure(figsize=(15,4))

# Temperature Graph
plt.subplot(1,3,1)
plt.plot(patients, temperature, marker='o')
plt.axhline(38,color='r',linestyle='--')
plt.title("Temperature")
plt.xticks(rotation=90)

# Heart Rate Graph
plt.subplot(1,3,2)
plt.plot(patients, heart_rate, marker='o')
plt.axhline(100,color='r',linestyle='--')
plt.title("Heart Rate")
plt.xticks(rotation=90)

# SpO2 Graph
plt.subplot(1,3,3)
plt.plot(patients, spo2, marker='o')
plt.axhline(95,color='r',linestyle='--')
plt.title("SpO₂")
plt.xticks(rotation=90)

plt.tight_layout()
```

```
    plt.show()

    # Wait 5 seconds
    time.sleep(5)

print("\nMonitoring Completed Successfully")

print("\nFinal Summary")
display(df)

print("\nTotal Patients :", total_patients)
print("Normal Patients :", len(df[df['Status']=='NORMAL']))
print("Emergency Patients :", len(df[df['Status']=='EMERGENCY']))
```

A Smart Hospital Monitoring System continuously observes the health condition of patients by collecting their vital signs such as body temperature, heart rate, and oxygen saturation. In real hospitals, these values are obtained using medical sensors connected through IoT or IoE devices. Since physical hardware is not used in this experiment, the patient data are simulated using randomly generated values that resemble real medical readings.

The system processes one patient every five seconds and updates the monitoring dashboard dynamically. After every update, the patient's vital signs are analysed against predefined threshold values. If any vital sign falls outside the normal range, the patient is classified as EMERGENCY; otherwise, the patient is considered NORMAL.

The generated information is displayed in a live table and continuously updated graphs, allowing healthcare staff to monitor patient conditions effectively. This simulation demonstrates the working principle of a real-time patient monitoring system without requiring physical medical devices.

Algorithm

Start the program.

Import all required Python libraries.

Set the total number of patients to 25.

Set the monitoring interval to 5 seconds.

Create empty lists to store patient information.

Generate random Temperature, Heart Rate, and SpO_2 values.

Compare each patient's values with predefined threshold limits.

Assign the patient status as NORMAL or EMERGENCY.

Store the generated data.

Display the live patient monitoring table.

Display emergency patients separately.

Update the live graphs.

Wait for five seconds.

Repeat until all 25 patients are monitored.

Display "Monitoring Completed Successfully".

Stop the program.

Flowchart

```
        START
          |
          ▼
  Import Required Libraries
          |
          ▼
  Initialize Monitoring System
          |
```

▼

Set Total Patients = 25

|

▼

Generate Patient Data

|

▼

Temperature, Heart Rate, SpO_2

|

▼

Compare with Threshold Values

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▼ ▼

NORMAL EMERGENCY

| |

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▼

Store Patient Information

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▼

Display Live Monitoring Table

|

▼

```
      Update Live Graphs
                 |
                 ▼
   Wait 5 Seconds for Next Patient
                 |
                 ▼
    More Patients Remaining?
          |         |
         Yes        No
          |         |
          ▼         ▼
   Generate Next     Display Final
   Patient Data      Summary
          |
          └──────────────────────────▶
                 ▼
                END
```

Threshold Values Used

Parameter	Normal Range	Emergency Condition
Temperature	36.5°C – 37.5°C	>38°C or <35°C
Heart Rate	60–100 BPM	<60 or >100 BPM
SpO_2	95–100%	<95%

Program Description

The Python program simulates a smart hospital environment where 25 patients are monitored one after another. For every patient, the system generates body temperature, heart rate, and

SpO_2 values. These values are stored in a DataFrame and displayed in a live monitoring table. Simultaneously, the system updates graphical charts representing the patient's health parameters. Based on predefined threshold values, each patient is automatically classified as either NORMAL or EMERGENCY. Whenever an emergency patient is detected, an alert message and the patient's details are immediately displayed. The simulation continues until all patients have been monitored.

Input

Total Patients = 25

Monitoring Interval = 5 seconds

Simulated Temperature

Simulated Heart Rate

Simulated SpO_2

Output

Live Patient Monitoring Table

Temperature Graph

Heart Rate Graph

SpO_2 Graph

Emergency Alert Table

Final Monitoring Summary

Sample Output

Patient	Temperature	Heart Rate	SpO_2	Status
P01	36.9°C	78 BPM	98%	NORMAL
P02	39.1°C	112 BPM	92%	EMERGENCY
P03	37.0°C	80 BPM	97%	NORMAL
P04	36.8°C	75 BPM	99%	NORMAL
P05	38.6°C	110 BPM	93%	EMERGENCY

Observations

Patient vital signs are generated every five seconds.

The monitoring table updates automatically after each patient.

Graphs display real-time changes in Temperature, Heart Rate, and SpO_2.

Patients exceeding threshold limits are classified as EMERGENCY.

Emergency alerts are displayed immediately.

The simulation successfully monitors all 25 patients.

Applications

Smart hospitals

Patient monitoring systems

Intensive Care Units (ICU)

Telemedicine

Remote healthcare monitoring

Medical training and simulation

Healthcare data visualization

IoT-based healthcare research

Advantages

Real-time patient monitoring

Automatic emergency detection

Live graphical visualization

Easy to understand

Cost-effective simulation

No physical sensors required

Suitable for academic laboratories

Can be extended to real IoT devices

Limitations

Uses simulated patient data.

No real medical sensors are connected.

Does not send notifications to doctors.

Internet connectivity is not required.

Suitable only for educational demonstration.

Result

The Smart Hospital Live Patient Monitoring System was successfully simulated using Python in Google Colab. The system monitored 25 simulated patients by generating body temperature, heart rate, and SpO_2 values at five-second intervals. The patient information was displayed through a live monitoring table and continuously updated graphs. The system accurately classified patients as NORMAL or EMERGENCY based on predefined threshold values and displayed emergency alerts whenever abnormal conditions were detected. Thus, the experiment successfully demonstrated real-time patient monitoring, live data visualization, automated health status classification, and emergency detection in a simulated smart healthcare environment.

Patient	**Temperature (°C)**	**Heart Rate (BPM)**	**SpO_2 (%)**	**Status**	
0	P01	36.1	113	96	EMERGENCY
1	P02	38.2	80	100	EMERGENCY
2	P03	36.6	68	94	EMERGENCY
3	P04	37.5	103	90	EMERGENCY
4	P05	34.7	60	94	EMERGENCY
5	P06	35.1	55	97	EMERGENCY
6	P07	38.8	90	90	EMERGENCY

Patient	Temperature (°C)	Heart Rate (BPM)	SpO_2 (%)	Status	
7	P08	35.8	118	95	EMERGENCY
8	P09	38.4	88	96	EMERGENCY
9	P10	36.8	82	96	NORMAL
10	P11	36.0	103	90	EMERGENCY
11	P12	36.6	113	90	EMERGENCY
12	P13	38.3	96	90	EMERGENCY
13	P14	39.4	106	96	EMERGENCY
14	P15	35.0	92	90	EMERGENCY
15	P16	35.5	72	97	NORMAL
16	P17	36.7	51	90	EMERGENCY
17	P18	37.6	57	95	EMERGENCY
18	P19	37.0	72	91	EMERGENCY
19	P20	39.3	117	92	EMERGENCY
20	P21	37.2	56	93	EMERGENCY
21	P22	38.3	53	90	EMERGENCY
22	P23	36.9	93	90	EMERGENCY

Patient	Temperature (°C)	Heart Rate (BPM)	SpO_2 (%)	Status	
23	P24	39.5	52	96	EMERGENCY
24	P25	38.6	76	93	EMERGENCY

Emergency Patients

	Patient	Temperature (°C)	Heart Rate (BPM)	SpO_2 (%)	Status
0	P01	36.1	113	96	EMERGENCY
1	P02	38.2	80	100	EMERGENCY
2	P03	36.6	68	94	EMERGENCY
3	P04	37.5	103	90	EMERGENCY
4	P05	34.7	60	94	EMERGENCY
5	P06	35.1	55	97	EMERGENCY
6	P07	38.8	90	90	EMERGENCY
7	P08	35.8	118	95	EMERGENCY
8	P09	38.4	88	96	EMERGENCY
10	P11	36.0	103	90	EMERGENCY
11	P12	36.6	113	90	EMERGENCY
12	P13	38.3	96	90	EMERGENCY

	Patient	Temperature (°C)	Heart Rate (BPM)	SpO$_2$ (%)	Status
13	P14	39.4	106	96	EMERGENCY
14	P15	35.0	92	90	EMERGENCY
16	P17	36.7	51	90	EMERGENCY
17	P18	37.6	57	95	EMERGENCY
18	P19	37.0	72	91	EMERGENCY
19	P20	39.3	117	92	EMERGENCY
20	P21	37.2	56	93	EMERGENCY
21	P22	38.3	53	90	EMERGENCY
22	P23	36.9	93	90	EMERGENCY
23	P24	39.5	52	96	EMERGENCY
24	P25	38.6	76	93	EMERGENCY